

# Lecture 03 - Morphology

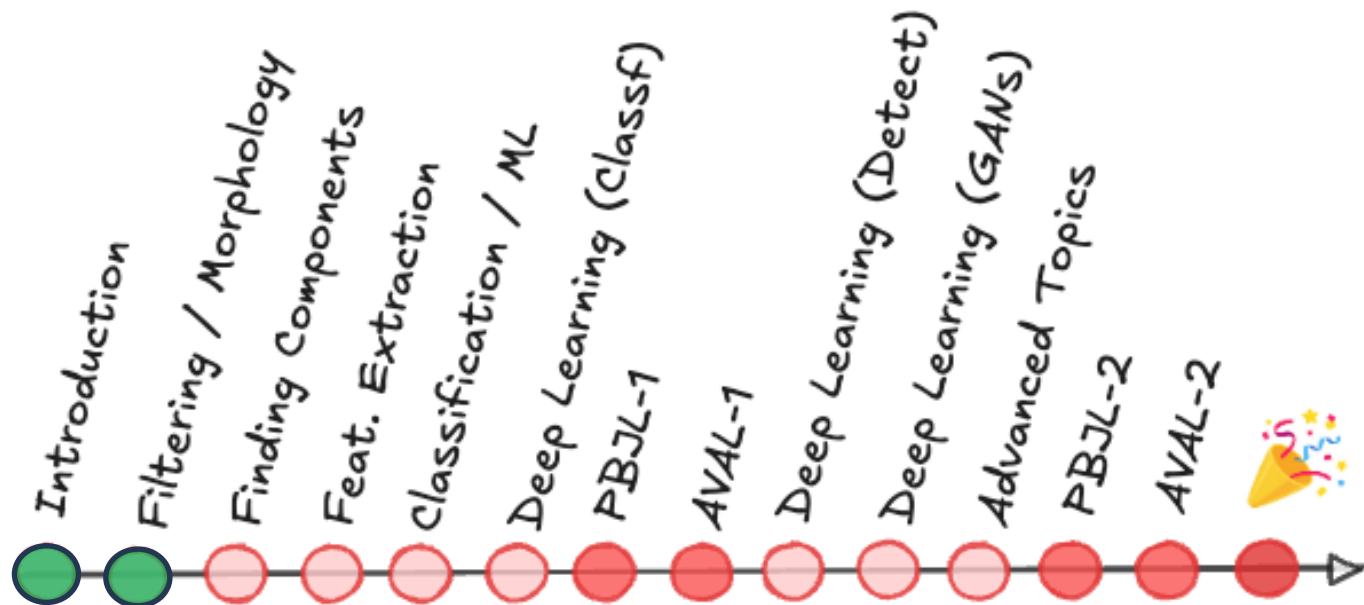
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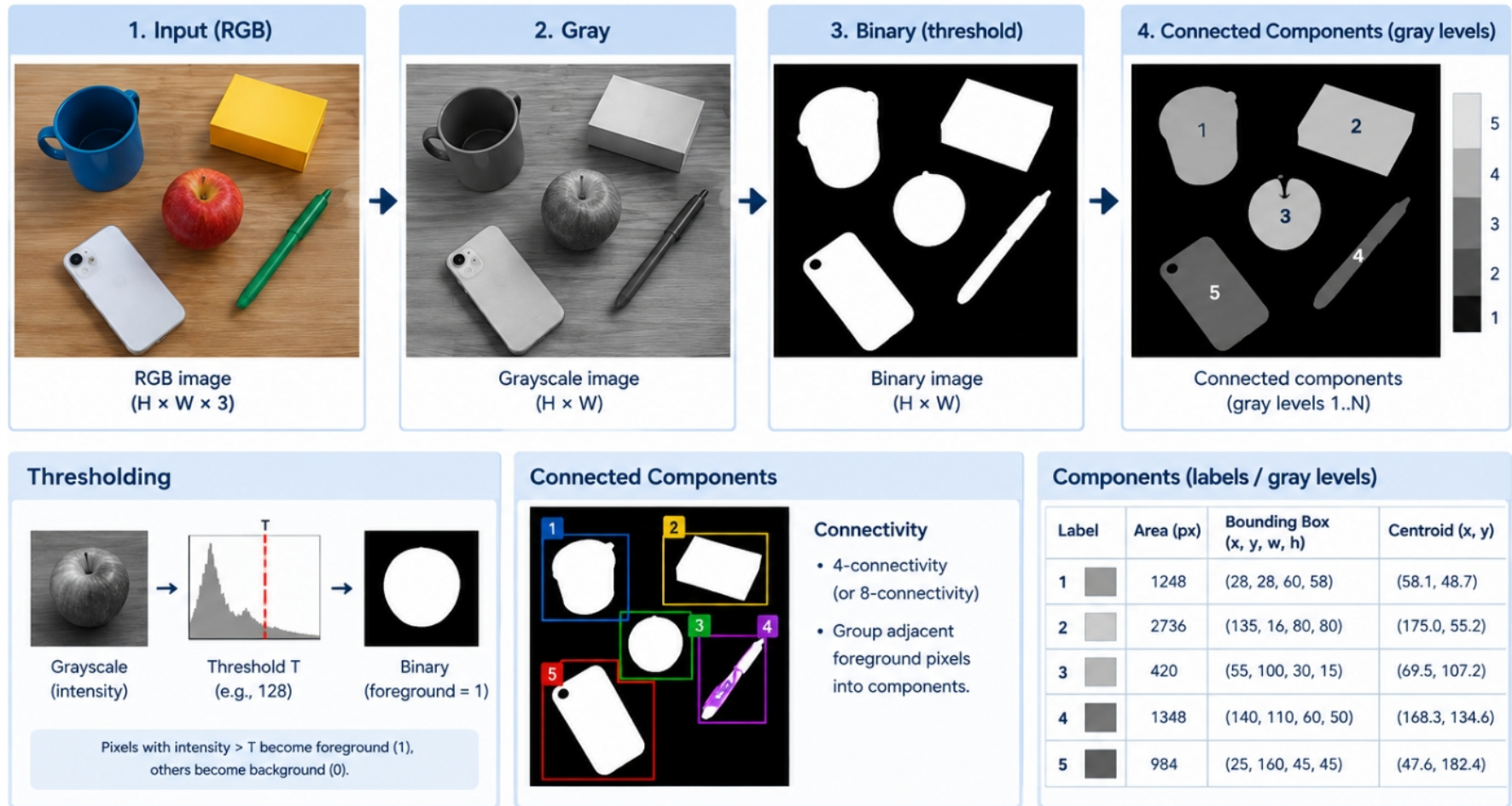
[aghochuli@ppgia.pucpr.br](mailto:aghochuli@ppgia.pucpr.br)

# Topics

- Discussion of Practice 02
- Image Binarization
- Mathematical Morphology
  - Structuring Element (or Kernel)
  - Erode / Dilate
  - Open / Close
  - Gradients
  - Watershed
- Practice



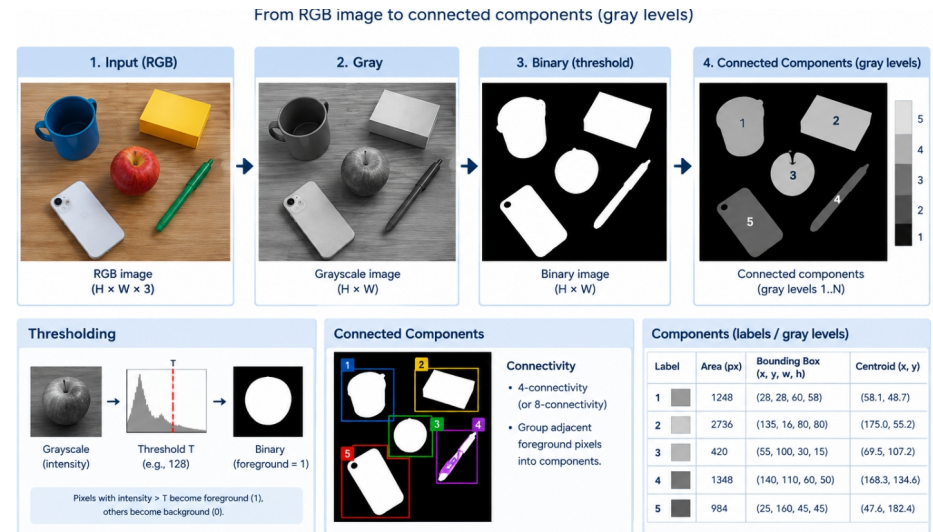
# Image Binarization



# Image Binarization

## Why Binarize the Image?

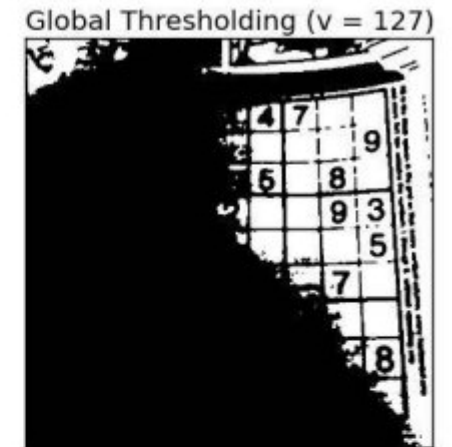
- Separate foreground from background
  - Converts the image into two classes: object (1) and background (0).
- Simplify the image representation
  - RGB:  $H \times W \times 3$
  - Gray:  $H \times W$ , many intensity values
  - Binary:  $H \times W$ , only 0 or 1
- Enable object/component analysis
  - Connected-component algorithms operate on foreground pixels and group spatially connected pixels into individual objects.
  - Reduce irrelevant information
  - Once segmentation is defined, exact color/intensity values are no longer necessary for identifying components.



# Image Binarization – Threshold Selection

Defining the threshold value  $T$  is a key step

- $B(x,y) = \{1, 0, I(x,y) \geq T | I(x,y) < T\}$ 
  - Threshold selection is critical (20, 50, 127, 250?)
  - Too low → background and noise become foreground
  - Too high → object pixels and details are lost
  - Appropriate  $T$  → good foreground–background separation
- Common approaches
  - Global Thresholding: a single  $T$  for the entire image
  - Otsu's Method: automatically estimates an optimal global  $T$
  - Adaptive Thresholding:  $T$  varies according to local image regions

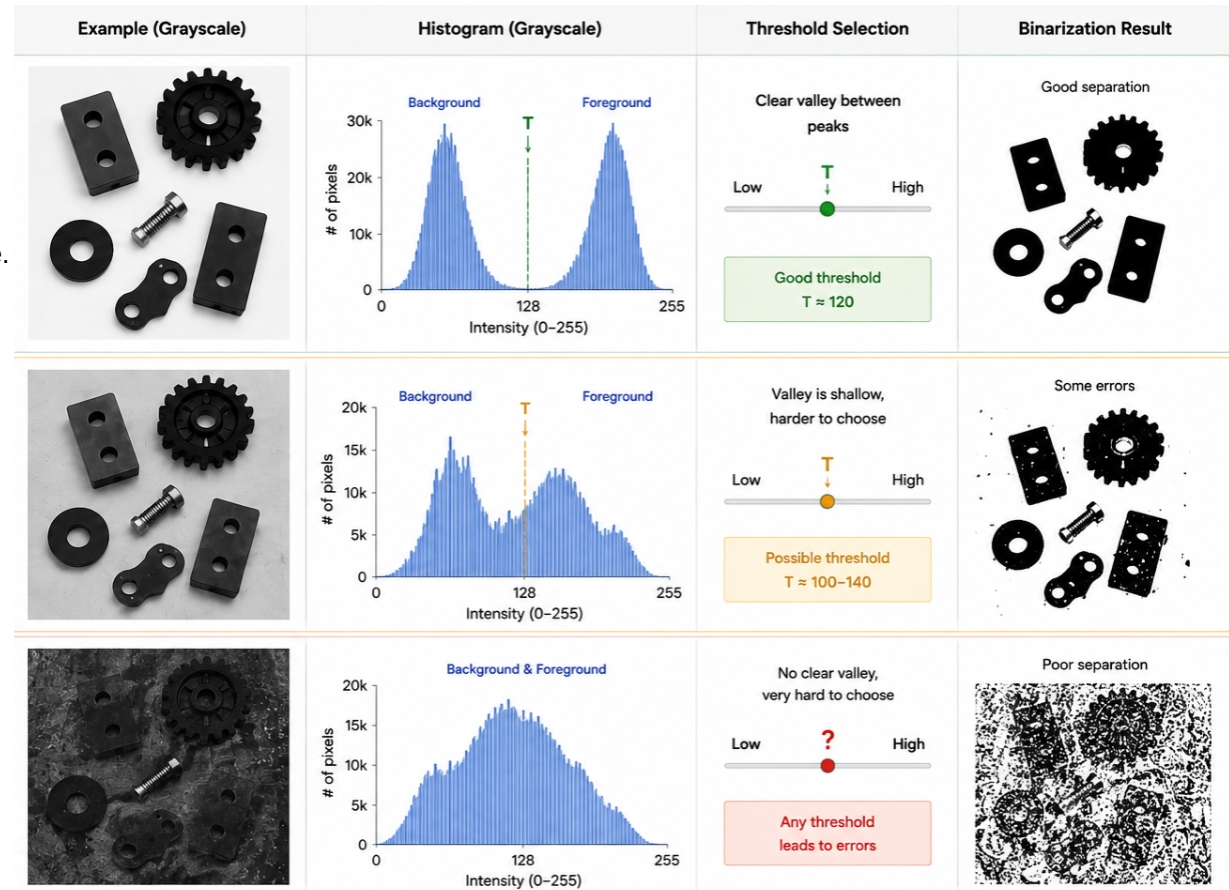


image

# Image Binarization – Threshold Selection

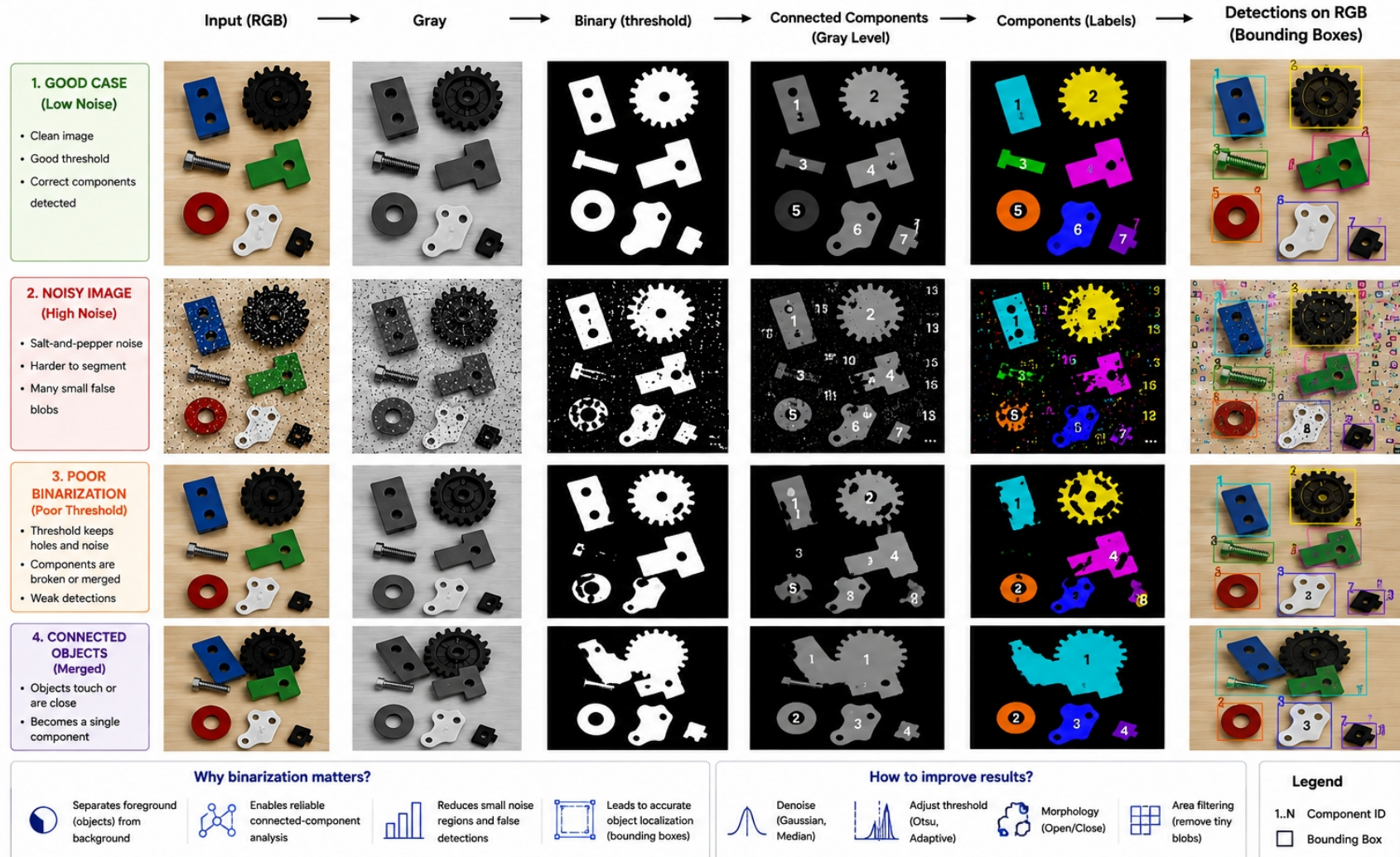
Defining the threshold value  $T$  is a key step

- Image Histogram
  - Represents the distribution of pixel intensities in an image.
  - Y-axis: number (or frequency) of pixels at each intensity.
  - Provides a global description of image appearance:
    - Brightness
    - Contrast
    - Intensity distribution
  - Useful for binarization
  - Peaks may represent different image regions/classes.
- A threshold  $T$  can be selected near the valley between the peaks.



# Image Binarization

How noise and binarization quality impact connected components and detections



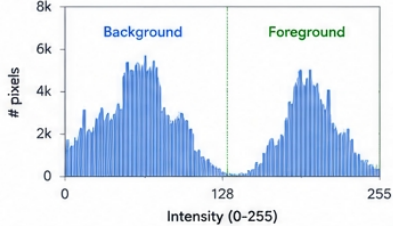
# Image Binarization

**Convert a grayscale image to binary (foreground = white, background = black)**

**Input**



**Histogram (grayscale)**



The histogram shows two main peaks (background and foreground) with a valley between them. The threshold should be chosen in the valley.

**How to Choose the Threshold (T)?**

- Visual inspection (quick)**  
Move the threshold and choose the value that best separates foreground from background.
- Histogram valley**  
Choose T in the valley between the background and foreground peaks.
- Automatic methods**  
Use an automatic thresholding method (e.g., Otsu, Triangle, Sauvola) that finds the best T based on image statistics.

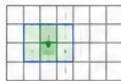
**Visual threshold selection**

T = 60  


T = 120 (✓)  


T = 180  


Choose T that best separates objects (white) from background (black) without losing details or adding too much noise.

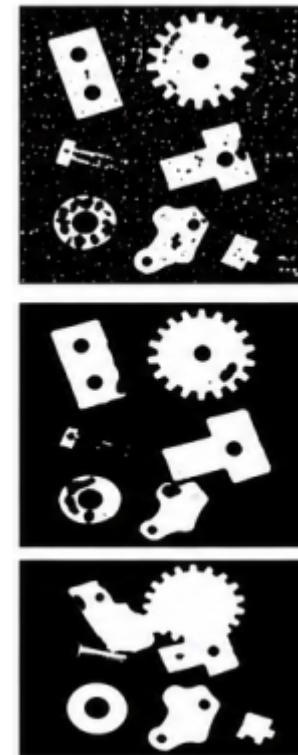
| Binarization Method | 1. Global Threshold (Manual)                                                                                                                                     | 2. Otsu (Global)                                                                                                                                              | 3. Adaptive Mean (Local)                                                                                                                                                                    | 4. Adaptive Gaussian (Local)                                                                                                                                                       |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| How it works        | Uses a single threshold $T$ for the entire image.<br><br>$\text{pixel}(x,y) \geq T \rightarrow \text{white}$<br>$\text{pixel}(x,y) < T \rightarrow \text{black}$ | Automatically finds $T$ that maximizes the between-class variance.<br><br> | Computes the mean in a local neighborhood and uses it as the threshold.<br><br><br>$T = \text{mean} - C$ | Like Adaptive Mean, but uses a Gaussian-weighted sum.<br><br><br>$T = \text{weighted mean} - C$ |
| Best for            | Images with uniform lighting and clear contrast.                                                                                                                 | Images with bimodal histograms (two peaks).                                                                                                                   | Images with uneven lighting (shadows, gradients).                                                                                                                                           | Images with uneven lighting and smooth variations.                                                                                                                                 |
| Result (Example)    |                                                                               |                                                                           |                                                                                                        |                                                                                               |
| Pros                | <ul style="list-style-type: none"> <li>✓ Simple and fast</li> <li>✓ Full control</li> </ul>                                                                      | <ul style="list-style-type: none"> <li>✓ Automatic</li> <li>✓ Works well for bimodal histograms</li> </ul>                                                    | <ul style="list-style-type: none"> <li>✓ Handles uneven lighting</li> </ul>                                                                                                                 | <ul style="list-style-type: none"> <li>✓ Handles uneven lighting</li> <li>✓ Less blocky than mean</li> </ul>                                                                       |
| Cons                | <ul style="list-style-type: none"> <li>✗ Sensitive to lighting</li> <li>✗ Needs manual tuning</li> </ul>                                                         | <ul style="list-style-type: none"> <li>✗ May fail with unimodal histograms</li> </ul>                                                                         | <ul style="list-style-type: none"> <li>✗ May produce blocky artifacts</li> <li>✗ Sensitive to window size</li> </ul>                                                                        | <ul style="list-style-type: none"> <li>✗ Sensitive to window size and parameters</li> </ul>                                                                                        |
| OpenCV Function     | <code>cv2.threshold()</code><br>(THRESH_BINARY)                                                                                                                  | <code>cv2.threshold()</code><br>(THRESH_OTSU)                                                                                                                 | <code>cv2.adaptiveThreshold()</code><br>(ADAPTIVE_THRESH_MEAN_C)                                                                                                                            | <code>cv2.adaptiveThreshold()</code><br>(ADAPTIVE_THRESH_GAUSSIAN_C)                                                                                                               |

**Tips:** Start with Otsu for automatic binarization. If lighting is uneven, try Adaptive Gaussian or Sauvola. Always validate the result visually for your specific application.

$m = \text{mean}$ ,  $s = \text{standard deviation}$ ,  $k = \text{constant}$  (e.g., 0.2–0.5),  $R = \text{dynamic range}$  (e.g., 128)

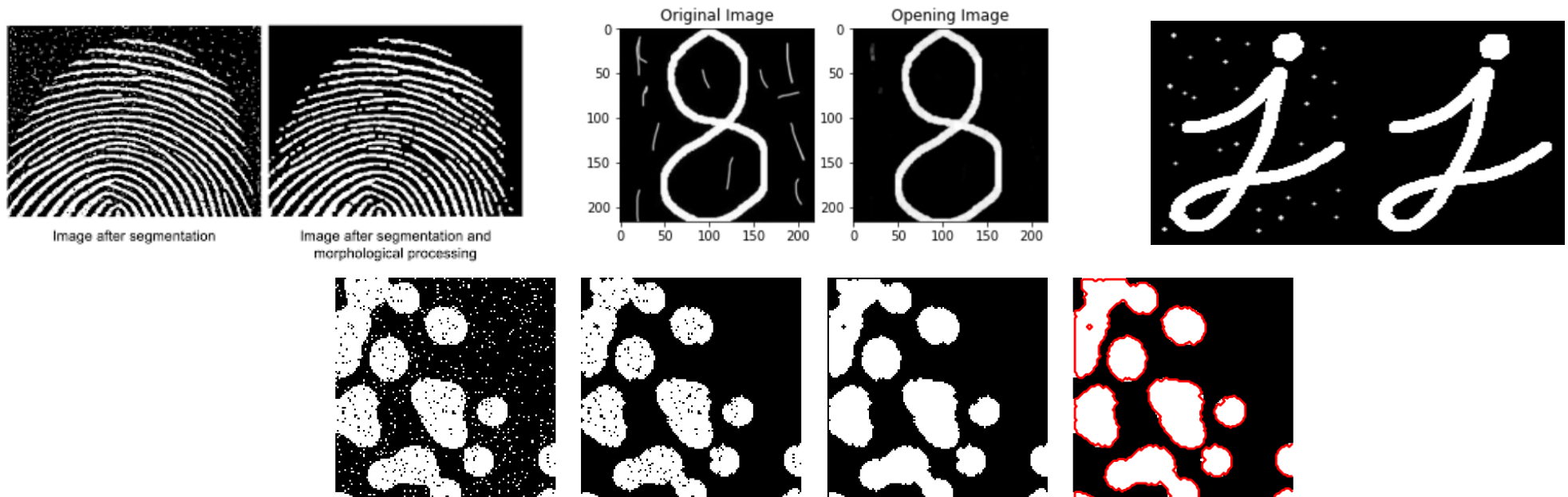
# Image Binarization

**What If Binarization Produces Noisy or Fragmented Objects?**



# Mathematical Morphology

- Morphology (Nature): Branch of Biology that study the form and structure of animals and plants
- Morphology (Image Processing): Mathematical operations to extract image components based on pixel neighborhood.
- Increase/Decrease Objects Size, Reduce Noise, Closing or Open



# Structuring Element (Kernel)

- Defines the shape of the structure to be applied
- The structure is slid through the image
- Erosion or Dilate operations are applied
- The origin determines the pixel be changed

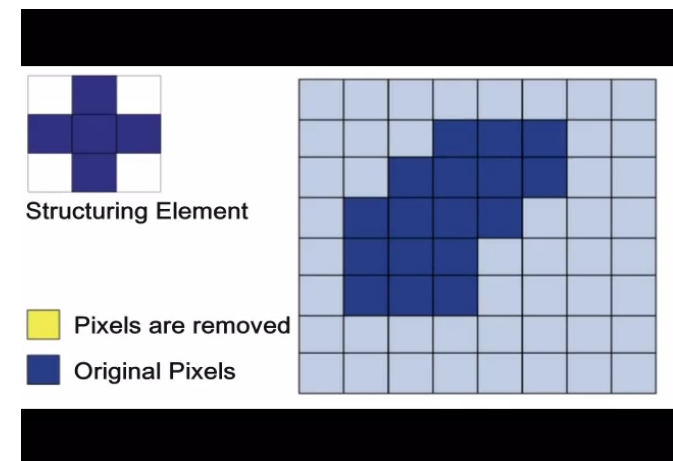
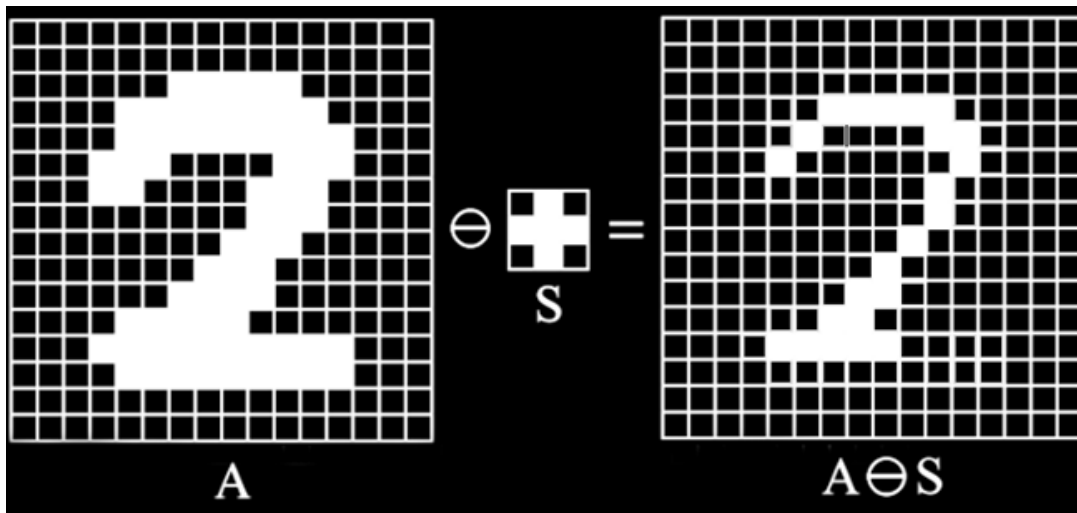
|   |   |   |
|---|---|---|
| 1 | 1 | 1 |
| 1 | 1 | 1 |
| 1 | 1 | 1 |

|   |   |   |   |   |   |   |
|---|---|---|---|---|---|---|
|   |   | 1 | 1 | 1 |   |   |
|   | 1 | 1 | 1 | 1 | 1 |   |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| 1 | 1 | 1 | 1 | 1 | 1 | 1 |
|   | 1 | 1 | 1 | 1 | 1 |   |
|   |   | 1 | 1 | 1 |   |   |

|   |   |   |
|---|---|---|
|   | 1 |   |
| 1 | 1 | 1 |
|   | 1 |   |

# Erosion

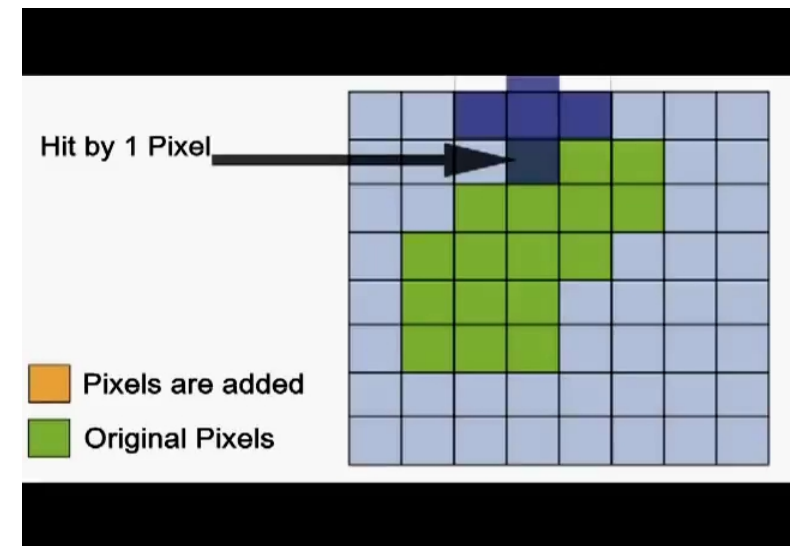
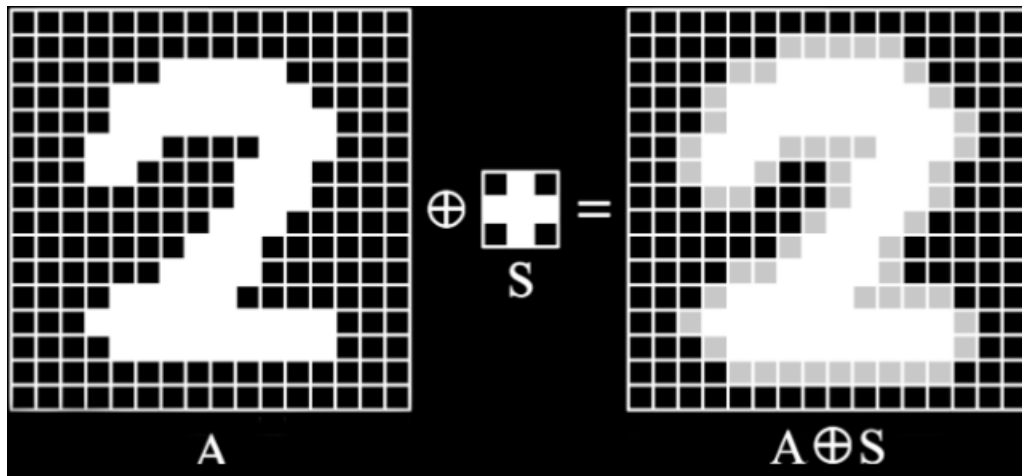
- Moves one's to zero's
- Binary 'AND' Operation
- Reduces Noises and Contours



<https://www.youtube.com/watch?v=fmyE7DiaIYQ>

# Dilation

- Moves zero's to one's
- Binary 'OR' Operation
- Increases Objects, connects contours, and fill holes.



<https://www.youtube.com/watch?v=xO3ED27rMHs>

# Opening and Closing

- Combined morphological operations that preserve the shape and size of large objects in the image

- Opening: Erode + Dilate
  - Removes small objects, noises, and thin lines



- Closing: Dilate + Erode
  - Fill small holes and connect segmented contours

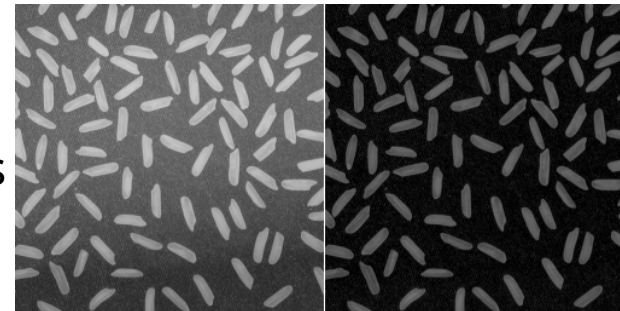


# Others Morphological Operations

- Gradient: Dilate - Erosion
  - Outline de Object



- TopHat: Open – Original Image
  - Reduce the effect of brightness changes
  - Isolates brightness objects



# Let's code!

- [Link: Morphology Operators](#)